

# Conjecture 9.5

Unrefereed candidate proof by GPT-6 Astra

Writeup: gpt-6-astra (single model pass)

Catalog id 2211.01032\_\_03 — generated 2026-09-16

**UNREFEREED MODEL OUTPUT.** This document was selected solely because GPT-6 Astra labelled its own result would\_publish: true and returned proved or disproved. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim below as unverified.

## Summary and provenance

|                        |                                                                                                                                  |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Catalog id             | 2211.01032__03                                                                                                                   |
| Catalog entry          | <a href="#">Conjecture 9.5</a>                                                                                                   |
| Source corpus          | arxiv                                                                                                                            |
| Campaign leg           | selected second attempts (attacks_retry)                                                                                         |
| Source paper / entry   | Random Embeddings of Graphs: The Expected Number of Faces in Most Graph...                                                       |
| Model verdict          | <b>proved</b> (confidence: high)                                                                                                 |
| Model's one-line claim | A fixed-vertex decomposition proves $E[F_n] \leq \ln n + O(1)$ , with the stronger upper bound $\ln n + 1/4 + \gamma/2 + o(1)$ . |
| Model                  | gpt-6-astra, reasoning effort max, mode=pro, flex service tier                                                                   |
| Original artifact      | attacks_retry/2211.01032__03/output.md                                                                                           |
| Independent review     | <b>None</b>                                                                                                                      |
| Model's caveats        | The proof uses exactly the stated uniform-rotation, independent-fair-signature model; literature priority has not been checked.  |

## Problem statement

No extracted statement was stored in the artifact metadata.

## Astra writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

### A fixed-vertex proof

Let  $F_n$  denote the number of faces in the random signed-rotation embedding of  $K_n$  specified in the question.

The key is to separate faces meeting one fixed vertex from faces avoiding it. Each class has expected size at most  $\frac{1}{2} \ln n + O(1)$ . For the avoiding faces, an unvisited vertex supplies an escape option at every exploration step. This controls local congestion and removes the long-face obstruction in the previous attempt.

The flag-matching representation from that attempt is valid and is justified below. Its short-face enumeration is not needed.

### Theorem

As  $n \rightarrow \infty$ ,

$$\mathbb{E}F_n \leq \ln n + \frac{1}{4} + \frac{\gamma}{2} + O\left(\frac{(\ln n)^2}{n}\right),$$

where  $\gamma$  is Euler's constant. In particular,

$$\mathbb{E}F_n \leq \ln n + O(1).$$

All logarithms below are natural unless a base is indicated.

## 1. The matching model and its exposure rule

Put  $d = n - 1$ . For each incidence of an edge  $uv$  at  $u$ , introduce two flags

$$(u, v, 0), \quad (u, v, 1).$$

Thus there are  $2d$  flags at each vertex and  $2n(n - 1)$  flags altogether.

Use the following matchings:

- $J_u$  pairs the two flags of each incidence at  $u$ .
- $A$  pairs corresponding flags at opposite ends of an edge:

$$(u, v, s) \longleftrightarrow (v, u, s).$$

- $V_u$  is a matching on the flags at  $u$ , describing the arcs between consecutive edge attachments around its vertex-disc.

The boundary components of the resulting ribbon graph are exactly the connected components of

$$A \cup V, \quad V = \bigcup_u V_u.$$

A face of length  $k$  contains  $k$  matching edges from  $A$ ,  $k$  from  $V$ , and  $2k$  flags. Consequently,

$$\sum_{\text{faces } f} |f| = n(n - 1).$$

### Distribution of the local matchings

Let

$$\mathcal{H}_d = \{W : W \text{ is a perfect matching and } J_u \cup W \text{ is one alternating } 2d\text{-cycle}\}.$$

In the stated random-embedding model, we may take the  $V_u$  to be independent and uniform on  $\mathcal{H}_d$ , with  $A$  fixed as above.

Here is a verification. Choose independent fair bits  $b_{u,e}$  at all incidences, and define the signature of  $e = uv$  by

$$\varepsilon_e = b_{u,e} \oplus b_{v,e},$$

with an immaterial fixed adjustment if the opposite sign convention is used. These signatures are independent and fair. Relabelling the flags at incidence  $(u, e)$  using  $b_{u,e}$  makes the edge matching canonical.

At a vertex, a cyclic order and the  $d$  incidence bits specify an element of  $\mathcal{H}_d$ . Every element has exactly two such representations, corresponding to the two directions around its alternating cycle. Hence

$$|\mathcal{H}_d| = 2^{d-1}(d - 1)!,$$

and the local matchings are independent and uniform, as asserted.

**Local exposure lemma** Suppose  $r < d$  pairs of  $V_u$  have been exposed, forming a matching  $Q_u$ .

Because  $J_u \cup V_u$  is one cycle,  $J_u \cup Q_u$  consists of  $d - r$  alternating paths. Its endpoints are exactly the flags whose  $V_u$ -mates are still unexposed.

If  $r \leq d - 2$ , then, for any such flag  $x$ , its mate is uniform among

$$2(d - r - 1) \quad (1)$$

possible flags: all the other free flags except the opposite endpoint of the same  $J_u \cup Q_u$ -path. If  $r = d - 1$ , the last pair is forced.

Indeed, contracting the alternating paths reduces the conditional completion problem to a uniform element of  $\mathcal{H}_{d-r}$ . Joining two different paths is allowed, while closing a proper local cycle is forbidden. All allowed partners have the same number of completions.

This rule remains valid under adaptive exposure using only previously revealed information.

## 2. Faces meeting a fixed vertex

Fix a vertex  $o$ . Write

$$F_n = F_n^{\text{hit}} + F_n^{\text{avoid}},$$

where the two terms count faces meeting and avoiding  $o$ , respectively.

Expose all local matchings except  $V_o$ . The components of the exposed boundary graph are:

- cycles, which are precisely the faces avoiding  $o$ ;
- paths whose endpoints are flags at  $o$ .

The latter paths induce a perfect matching  $P$  on the  $2d$  flags at  $o$ . Conditional on this exposure,  $V_o$  is still uniform on  $\mathcal{H}_d$ , and the faces meeting  $o$  correspond exactly to the components of  $P \cup V_o$ .

Expose  $V_o$  pair by pair. After  $r \leq d - 2$  pairs have been exposed, select any free flag  $x$ . At most one possible mate closes a component of  $P$  together with the exposed pairs: the opposite endpoint of the current  $P$ -path containing  $x$ . By (1), the conditional probability of completing a component is at most

$$\frac{1}{2(d - r - 1)}.$$

The final forced pair completes at most one component. Every component is counted when its last pair is exposed. Therefore

$$\mathbb{E} F_n^{\text{hit}} \leq 1 + \frac{1}{2} H_{d-1} = 1 + \frac{1}{2} H_{n-2}. \quad (2)$$

The remainder of the proof bounds  $F_n^{\text{avoid}}$ .

## 3. Exploring one face while avoiding $o$

We now use a different exposure order, analyzing one starting flag at a time and revealing no other faces.

Choose a flag  $a_0$  on an edge not incident with  $o$ , and put

$$b_0 = A(a_0).$$

Start with the  $A$ -edge  $a_0 b_0$ , treating  $a_0$  as the active flag and  $b_0$  as a fixed terminal.

At each step:

1. reveal the  $V$ -mate  $z$  of the active flag;

2. if  $z = b_0$ , the face closes;
3. otherwise, make  $A(z)$  the new active flag;
4. stop if the new active flag is at  $o$ .

Until closure, all exposed  $V$ -pairs lie on one path with endpoints the active flag and  $b_0$ . All other components of  $A$  together with these exposed pairs are isolated  $A$ -edges. Thus every query is to a previously unexposed mate.

If the trace closes after  $k$  queries, its face has length  $k$ .

**The escape probability** Suppose the process is alive, its active flag is at  $u \neq o$ , and  $r_u$  pairs have been exposed at  $u$ . Neither flag of incidence  $(u, uo)$  has been used. Otherwise the trace would already have visited  $o$ . Consequently those two flags form an isolated  $J_u$ -path, different from the path containing the active flag. Both are allowed choices in the local exposure lemma.

In particular, there are at least two unfinished local paths, so the forced-last-pair case cannot occur while the trace avoids  $o$ . The conditional probability of moving next to  $o$  is exactly

$$p_o = \frac{2}{2(d - r_u - 1)} = \frac{1}{n - 2 - r_u}. \quad (3)$$

For any other vertex  $w$ , there are at most two flags at  $u$  whose  $A$ -partners lie at  $w$ . Some might already be unavailable, and choosing the terminal may instead close the face. Thus the conditional probability  $p_w$  of making  $w$  the next active vertex satisfies

$$p_w \leq p_o. \quad (4)$$

Let  $T$  be the number of queries until closure or the first visit to  $o$ . Set

$$q = 1 - \frac{1}{n - 2}.$$

Equation (3) gives

$$\boxed{\Pr(T > t) \leq q^t.} \quad (5)$$

**A geometric bound on repeated visits** Fix  $w \neq o$ . While the trace is alive, let  $N_w(t)$  count its active visits to  $w$ , including the current active position and the initial position if applicable.

Define

$$M_t = \begin{cases} 2^{N_w(t)}, & t < T, \\ 0, & t \geq T. \end{cases}$$

Let  $p_{cl}$  denote the conditional probability of closing at the next query. While alive,

$$\mathbb{E}[M_{t+1} \mid \mathcal{F}_t] = M_t(1 - p_o - p_{cl} + p_w) \leq M_t$$

by (4). Hence  $M_t$  is a nonnegative supermartingale, with  $M_0 \leq 2$ .

Stop it on reaching  $L$  active visits to  $w$ , or on termination. There are at most  $n(n - 1)$  queries, so bounded stopping gives

$$\boxed{\Pr(N_w(t) \geq L \text{ for some } t < T) \leq 2^{1-L}.} \quad (6)$$

This controls local congestion throughout the entire trace, not merely up to a predetermined length.

#### 4. Bounding all faces avoiding $o$

For sufficiently large  $n$ , choose

$$R = \lceil 4 \log_2 n \rceil, \quad D = n - 2 - R > 0. \quad (7)$$

Call an avoiding face **good** if it uses at most  $R$  local pairs at every vertex, and **bad** otherwise.

Let  $S$  be the flags on edges not incident with  $o$ . Then

$$|S| = 4 \binom{n-1}{2} = 2(n-1)(n-2). \quad (8)$$

**4.1. Bad avoiding faces contribute  $o(1)$**  If the face of a starting flag avoids  $o$  and is bad, its trace reaches  $R + 1$  active visits to some vertex before terminating. By (6) and a union bound,

$$\Pr(\text{the starting flag lies in a bad avoiding face}) \leq (n-1)2^{-R}. \quad (9)$$

For a flag  $a$ , let  $L(a)$  be its face length. Every length- $k$  face has  $2k$  flags, so the number  $B$  of bad avoiding faces is exactly

$$B = \sum_{a \in S} \frac{\mathbf{1}_{\{a \text{ lies in a bad avoiding face}\}}}{2L(a)}.$$

Using  $L(a) \geq 1$ , (8), and (9),

$$\mathbb{E}B \leq (n-1)^2(n-2)2^{-R} \leq n^3 2^{-R} \leq \frac{1}{n}. \quad (10)$$

No independence between traces from different starting flags is used.

**4.2. Closing probabilities for good faces** Fix a starting flag, and let  $w_0$  be the vertex containing its terminal  $b_0$ .

For the trace to close at query  $k$ , it must:

- remain alive after query  $k - 2$ ;
- at query  $k - 1$ , move to  $w_0$ ;
- at query  $k$ , choose the particular terminal flag  $b_0$ .

When a revealed history has used at most  $R$  local pairs at every vertex, the exposure rule gives at least  $2D$  possible partners at every relevant query. Therefore:

- the probability of moving next to  $w_0$  is at most  $2/(2D) = 1/D$ ;
- once there, the probability of choosing  $b_0$  is at most  $1/(2D)$ .

These are bounds conditional on the revealed history, **not** on eventual goodness. To estimate a good-face event, discard continuations whose histories cease to satisfy the load restriction. Applying the two bounds successively, and then (5), gives, for  $k \geq 2$ ,

$$\Pr(\text{the starting flag lies in a good avoiding face of length } k) \leq \frac{q^{k-2}}{2D^2}. \quad (11)$$

This estimate applies to every possible face length.

**4.3. Summation over lengths** For  $n \geq 3$ , there are no faces of length one or two.

Length one is impossible because  $K_n$  has no loops. A length-two face would have to use both sides of one edge  $uv$ . At each endpoint, this would require  $V_u$  or  $V_v$  to pair the two flags of that same incidence. Such a pair would form a proper alternating cycle with  $J_u$  or  $J_v$ , contrary to membership in  $\mathcal{H}_d$  when  $d > 1$ .

Let  $G_k$  count good avoiding faces of length  $k$ . Rooting at flags, (8) and (11) give

$$\begin{aligned} \mathbb{E}G_k &= \frac{1}{2k} \sum_{a \in S} \Pr(a \text{ lies in such a face}) \\ &\leq \frac{(n-1)(n-2)}{2D^2} \frac{q^{k-2}}{k}. \end{aligned}$$

Together with (10),

$$\mathbb{E}F_n^{\text{avoid}} \leq \frac{(n-1)(n-2)}{2(n-2-R)^2} \sum_{k=3}^{\infty} \frac{q^{k-2}}{k} + \frac{1}{n}. \quad (12)$$

The series is explicit:

$$\begin{aligned} \sum_{k=3}^{\infty} \frac{q^{k-2}}{k} &= q^{-2} \left( -\ln(1-q) - q - \frac{q^2}{2} \right) \\ &= q^{-2} \left( \ln(n-2) - q - \frac{q^2}{2} \right) \\ &= \ln n - \frac{3}{2} + O\left(\frac{\ln n}{n}\right). \end{aligned} \quad (13)$$

Also, because  $R = O(\ln n)$ ,

$$\frac{(n-1)(n-2)}{2(n-2-R)^2} = \frac{1}{2} + O\left(\frac{\ln n}{n}\right). \quad (14)$$

Consequently,

$$\mathbb{E}F_n^{\text{avoid}} \leq \frac{1}{2} \ln n - \frac{3}{4} + O\left(\frac{(\ln n)^2}{n}\right). \quad (15)$$

## 5. Conclusion

From (2),

$$\mathbb{E}F_n^{\text{hit}} \leq 1 + \frac{1}{2}H_{n-2} = \frac{1}{2} \ln n + 1 + \frac{\gamma}{2} + O(1/n).$$

Adding (15) proves

$$\mathbb{E}F_n \leq \ln n + \frac{1}{4} + \frac{\gamma}{2} + O\left(\frac{(\ln n)^2}{n}\right).$$

This establishes the conjecture. The argument was stated for sufficiently large  $n$ ; finitely many smaller values are absorbed into the  $O(1)$  in the conjecture.

**If one conditions on non-orientability** The stipulated independent-signature model also produces orientable embeddings with a very small probability. If “non-orientable random embedding” is instead intended to mean conditioning on non-orientability, the conclusion remains unchanged.

For a connected graph, precisely  $2^{n-1}$  signatures are orientable: these are obtained by switching vertex orientations from an all-orientable signature. Thus, for  $K_n$ , the orientability probability is

$$p_n = 2^{n-1-\binom{n}{2}} = 2^{-(n-1)(n-2)/2}.$$

Since  $F_n \geq 0$ ,

$$\mathbb{E}[F_n \mid \text{non-orientable}] \leq \frac{\mathbb{E}F_n}{1 - p_n},$$

which preserves the displayed asymptotic upper bound.

The long-face gap is therefore closed: faces meeting the fixed vertex are controlled by a single local matching exposure, while avoiding faces admit both an exponential survival bound and a geometric local-congestion bound. No additional assumption about mixing or long facial walks is required.